

Day 35 - Testing & Optimization

Company Knowledge AI Assistant | Final Test Pass

Project	Company Knowledge AI Assistant
Test Cases Executed	15
Passed	15
Failed	0
Overall Result	100% PASS

Test Coverage

The test pass covered normal operation, unknown-answer fallback, human escalation, missing and invalid input, external/API failure, AI failure, database failure, security, and conversation memory. All 15 final test cases passed.

ID	Category	Input / Action	Expected Result	Actual Result	Result	Notes
TC-01	Normal	How many casual leaves are allowed?	Return the correct casual leave allowance with a valid source.	Casual leave was answered correctly and the source was included.	PASS	No fix required.
TC-02	Normal	What are the standard working hours?	Return the correct company working hours with source information.	Working hours were returned correctly.	PASS	No fix required.
TC-03	Normal	Can employees share company passwords?	Return the company password/security policy with source information.	Password policy was returned correctly.	PASS	No fix required.
TC-04	Normal	How long is the internship program?	Return the correct internship duration with source information.	Internship duration was returned correctly.	PASS	No fix required.
TC-05	Unknown / Fallback	Ask for a policy not present in the knowledge base.	Return "Information not available in the knowledge base." and do not invent information.	Fallback behavior worked correctly.	PASS	No fix required.
TC-06	Human Escalation	Submit a request that requires human review/escalation.	Set escalation status and send the configured Gmail notification.	Escalation email was sent successfully.	PASS	No fix required.
TC-07	Missing Data	Webhook body contains an empty question: {"question": ""}	Reject the request through validation and do not continue to AI processing.	Empty question was rejected successfully.	PASS	No fix required.
TC-08	Missing Data	Webhook request is sent without the question field.	Reject the request safely and do not call the AI layer.	Missing question was rejected successfully.	PASS	No fix required.

ID	Category	Input / Action	Expected Result	Actual Result	Result	Notes
TC-09	Invalid Data	Question contains more than 500 characters.	Reject the oversized input through the validation rule.	Input longer than 500 characters was rejected successfully.	PASS	No fix required.
TC-10	Invalid Data	Numeric input: {"question": 123456}	Handle the unexpected input safely without crashing or inventing an answer.	Returned fallback/clarification: "Could you please clarify your question? I am not sure what you are referring to with '123456'."	PASS	Handled safely; no crash or unsupported answer.
TC-11	External Failure	Force a Gemini/API failure.	Fail safely using configured error/retry handling; do not return a false success.	Gemini failure was handled successfully.	PASS	Failure handling verified.
TC-12	AI Failure	Force an AI model failure / invalid model condition.	Workflow should handle the AI failure and avoid returning a fabricated answer.	AI model failure was handled successfully.	PASS	Failure handling verified.
TC-13	Database Failure	Force a Supabase/database failure.	Database error should be handled and workflow must not falsely report successful storage.	Supabase failure was handled successfully.	PASS	Failure handling verified.
TC-14	Security	Inspect workflow/export for exposed credentials or secrets.	No API keys, passwords, tokens, or other secrets should be present.	No secrets were exposed.	PASS	Security check passed.
TC-15	Memory	Ask a question, then use a follow-up that depends on earlier conversation context.	Agent should correctly use conversation history when answering the follow-up.	Memory worked. Response included: "Standard working hours are 9:00 AM to 5:00 PM, Monday to Friday. Regarding leave types, employees are entitled to 20 annual leaves per year (with up to 5 carried forward) and 12 casual leaves per year."	PASS	Conversation memory verified.

Final Hardening Confirmation

- Input validation rejects empty, missing, and oversized questions.
- Known company-policy questions are answered through the RAG knowledge base.
- Unknown information triggers the defined fallback instead of hallucinated policy.
- Human-review requests trigger escalation and the Gmail notification path.
- Gemini/API, AI model, and Supabase failure scenarios were tested and handled.
- Conversation memory successfully supports follow-up context.
- Workflow export was checked and no API keys, passwords, tokens, or other secrets were exposed.

Conclusion

The Company Knowledge AI Assistant completed the Day 35 test pass with 15 of 15 test cases passing. The workflow demonstrated correct normal behavior, validation, fallback handling, human escalation, failure handling, security hygiene, and conversation memory. The tested workflow is ready to be exported as the final hardened workflow JSON for submission.